Carrie Lai

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Education

University of Southern California, Los Angeles, CA January 2023 - Spring 2025
Master of Science in Aerospace Engineering
Coursework: Space Physics, Computational Fluid & Plasma Dynamics, Combustion and Rocket Propulsion

University of Washington, Seattle, WA August 2017 - June 2021
Bachelor of Science in Aeronautical and Astronautical Engineering
Coursework: Orbital Mechanics, Fluid Mechanics, Aircraft & Spacecraft Dynamics and Control, CubeSat design

Honors

Selected Participant, National Science Foundation (NSF) Sponsored Program—USC Advanced Programming and Computational Fluid Dynamics, USC Center for Advanced Research Computing (Summer 2024)

- Developed advanced CFD programming skills in Python, MPI, CUDA, and Kokkos with USC High Performance Computing Cluster (HPC)
- Designed deep learning systems and applied neural networks for optimization in CFD

Professional & Research Experience

AI Trainer December 2025 - Present
Handshake AI Fellowship, Remote, Part-time

- Worked on high-volume, production-level datasets for clients to support large language models (LLM) training and performance
- Reviewed AI-generated content for accuracy, consistency, and guideline compliance and annotated quality-checked dataset used in model training and evaluation
- Identified edge cases and provided feedback to improve model behavior

Space Physics Research Intern, Space Weather Modeling Framework (SWMF) June 2025 - September 2025
Professor Shasha Zou, University of Michigan, Ann Arbor, MI

- Investigated fast earthward-propagating flows, Bursty Bulk Flows (BBFs ~10 mins, ~400,000 m/s), from magnetic reconnection in Earth's magnetotail, observed by Magnetospheric Multiscale Satellite
- Developed Python scripts to post-process BBFs dataset from NASA Pleiades Supercomputer and analyzed particle energization and heating processes as reported by satellite observations
- Collaborated with professors and post-doc researchers to optimize code performance, improve Maxwellian fitting methods, interpret physics results, contributing to NASA-funded space weather modeling research

Researcher, Combustion Physics Laboratory July 2024 - May 2025
Professor Paul Ronney, University of Southern California, Los Angeles, CA

- Utilized Starfish, a 2D gas/plasma codes developed by Particle in Cell LLC, to model plasma discharge region for the experimental design of the spherical plasma reactor
- Lead author; presented "*Exploring Plasma-Assisted Combustion in a Jet-Stirred Reactor Using Particle-In-Cell Simulation,*" at 14th U.S. National Combustion Meeting (USNCM 2025, Boston, MA)
- Contributor; presented "*Design and Restrictions on Spherical Jet-Stirred Plasma Reactor,*" at 2024 International Gaseous Electronics Conference (American Physical Society, San Diego, CA)

Fluid System Dev. Engineer January 2023 - August 2023
USC Liquid Rocket Propulsion Laboratory, Los Angeles, CA

- Designed and modeled flight vehicle components, including a mounting mechanisms and assemblies for a Composite Overwrapped Pressure Vessel (COPV) and associated propellant feed system

Selected Projects

Solid Rocket Motor Design August 2024 - December 2024
Professor David Reese, University of Southern California, Los Angeles, CA

- Developed custom 1D ballistics code to compute local pressure, velocity, and mass flux for thermal analysis of various motor structural components and selected hardware insulation material for motor case, domes and nozzle

Fluid/Plasma Modeling January 2024 - June 2024
Professor Lubos Brieda, University of Southern California, Los Angeles, CA

- Built 1D two-fluid Magnetohydrodynamics (MHD) codes to model plasma sheath dynamics, and evaluated numerical noise characteristics by comparing results with Particle-in-Cell (PIC) methods

- Simulated the ion thruster backflow plume using a C++ framework developed by Prof. Brieda, based on a legacy Fortran model originally created at NASA's Jet Propulsion Laboratory for the Deep Space 1 mission, to study how charge exchange ions drift back and interact with spacecraft surfaces
- Customized the simulation by implementing mass flow–based particle injection, reflective boundary conditions, and charge exchange (CEX) collision physics to analyze plume-surface interactions and assess potential damage in electric propulsion systems

Undergraduate Projects & Research

GNC Team, UW Aeronautics & Astronautics CubeSat Team September 2020 - June 2021

The Maratus Mission (6U CubeSat), Seattle, WA

- Led the design of controls system architecture, mission objectives, and Concept-of-Operations (CONOPs) for satellite guidance, navigation, and controls team and examined mission requirements, including attitude control, angular momentum management, and pointing accuracy
- Modeled environmental disturbance torques (e.g., atmospheric drag, gravity gradients, solar pressure) to assess impact on satellite dynamics and integrated actuators and sensors for the Attitude Determination and Control System (ADCS), ensuring effective torque and momentum management

Avionics Team September 2020 - April 2021

UW Design, Build and Fly Team, Seattle, WA

- Supported avionics design of a UAV, including motor selection, sensor integration, and flight control tuning, achieving 15-minute flights at 20 m/s with a 37-lb payload aircraft

Communication Team, UW Aeronautics & Astronautics CubeSat Team January 2020 - August 2020

SOC-I Mission (2U CubeSat), Seattle, WA

- Selected for NASA's CubeSat Launch Initiative (ELaNa-43), one of 18 CubeSat missions chosen nationwide (https://www.nasa.gov/missions/small-satellite-missions/nasas-elana-43-prepares-for-firefly-aerospace-launch/)
- Analyzed satellite ground pass statistics for various orbital altitudes and inclinations, including contact durations and frequencies, etc.

Research Assistant March 2018 - June 2019

UW Autonomous Flight System Lab, Seattle, WA

- Designed control systems of the Research Civil Aircraft Model (RCAM)

Skills

- Technical: C++, Python, MATLAB & Simulink, CAD (Siemens NX & SolidWorks), and COMSOL
- Additional: Teaching Assistant (Grader) - USC Department of Mathematics, School Ambassador - UW Husky Leadership Program, Student Coordinator - Starbucks (UW Housing & Food Service)

Other

- Authorized to work for any US employer (US Citizen)